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Natural Transformations

Definition

A **natural transformation** $\eta: F \Rightarrow G$ between functors $F, G: C \rightarrow D$ is:

Collection $\{\eta_X : F(X) \rightarrow G(X) \mid X \in C\}$ of morphisms in D

Naturality Axiom: For all $f: X \rightarrow Y$ in C , the diagram commutes:

$$\begin{array}{ccc}
 F(X) & \xrightarrow{\eta_X} & G(X) \\
 \downarrow F(f) & & \downarrow G(f) \\
 F(Y) & \xrightarrow{\eta_Y} & G(Y)
 \end{array}$$

This ensures η is “natural” - doesn’t depend on arbitrary choices.

Why Naturality?

Without naturality condition, just collection of morphisms.

With naturality: Forces compatibility with categorical structure.

Eilenberg-Mac Lane insight: Naturality formalizes “canonical” constructions.

15+ Examples

1. Double Dual (Finite-Dim Vect)

$\eta_V: V \rightarrow V^{**}$ sending v to evaluation at v Natural, but not iso for infinite-dimensional

2. Abelianization

Forgetful $\text{Grp} \rightarrow \text{Ab}$ composed with universal map Natural endomorphism

3. Fundamental Group

$\pi_1: \text{Top} \rightarrow \text{Grp}$ from basepoint Homotopy equivalence is natural transformation

4. Determinant

$\det: \text{GL}(n) \rightarrow k^*$ Natural between linear group and multiplicative

5. Homology Theories

Singular vs simplicial homology Natural isomorphic via Eilenberg-Zilber

6. Tensor-Hom Adjunction

Natural bijection between bilinear and linear maps

7. Yoneda Embedding

$y: C \rightarrow [C^{\text{op}}, \text{Set}]$ Fully faithful natural embedding

8. Unit/Counit of Adjunction

$\eta: \text{id} \Rightarrow G \circ F$ and $\varepsilon: F \circ G \Rightarrow \text{id}$ Natural transformations defining adjoints

9-15: Many more in algebra, topology, logic...

Composition of Natural Transformations

Vertical Composition

$\eta: F \Rightarrow G$ and $\mu: G \Rightarrow H$ give $(\mu \circ \eta): F \Rightarrow H$

Component: $(\mu \circ \eta)_X = \mu_X \circ \eta_X$

Horizontal Composition (Godement Product)

For $\eta: F \Rightarrow G$ ($C \rightarrow D$) and $\mu: F' \Rightarrow G'$ ($D \rightarrow E$):

$(\mu * \eta): F' \circ F \Rightarrow G' \circ G$

Component: Commutative square combining both

Interchange Law

$$(\mu_2 \circ \mu_1) * (\eta_2 \circ \eta_1) = (\mu_2 * \eta_2) \circ (\mu_1 * \eta_1)$$

Both compositions compatible!

Natural Isomorphism

Natural isomorphism: All components η_X are isomorphisms

$F \cong G$: Functors naturally isomorphic

Equivalence of categories: $F \circ G \cong \text{id}$ and $G \circ F \cong \text{id}$

Functor Categories

Natural transformations make $[C, D]$ into category: - Objects: Functors $C \rightarrow D$ - Morphisms: Natural transformations - Composition: Vertical composition

Key Theorems

- **Yoneda Lemma:** $\text{Nat}(\text{Hom}(-, A), F) \cong F(A)$
- **Mac Lane Coherence:** Associator/unitor natural isos satisfy pentagon
- **Adjoint Naturality:** Unit/counit are natural transformations

Common Mistakes

- ✗ “Just a collection of morphisms” ✓ Must satisfy naturality square
- ✗ “Natural = canonical” ✓ More precise: canonical constructions are natural
- ✗ “Horizontal composition = vertical” ✓ Different operations (Godement $\neq$ pointwise)

Study Guide

1. Master naturality square
2. Work through 5+ examples carefully
3. Prove vertical composition associative
4. Understand horizontal composition
5. Recognize natural isos in your field
6. Connect to adjoints and Yoneda

Questions

1. What does naturality square express?
2. Why call it “natural”?

3. How do compositions interact?
4. What makes Yoneda lemma natural?
5. How do natural transformations lift to categories?